

SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME-engines</i>
☒	Shut off control air supply
☒	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D06-18	Teeth on chain wheel, max. wear	3	mm
D06-19	Original length (chain pitch x 10 links)	889	mm
D06-20	10 links measurements + 1% of a tensioned chain = scrapping of chain	897.9	mm

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90651	44	Chain assembling tool
P90651	56	Chain disassembling tool
P91356		Lifting Tools, Etc.

Carry out the inspection as follows:

1. Make a general inspection for loose bolts and screws.
2. Inspect lube oil pipes for damage, and check jet nozzles for possible stoppages or deformations.
3. Examine the rubber track of the guideways for cracks or other damage.

Replace the guideway if bits have started to be “plucked out” of the rubber track.

4. Check the teeth of the chain wheels. If abnormal wear is found, take a measurement (see *Data*).

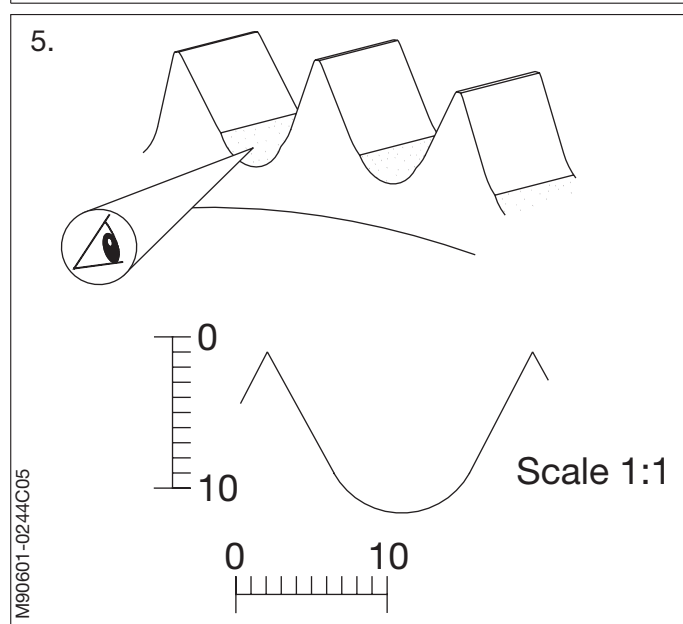
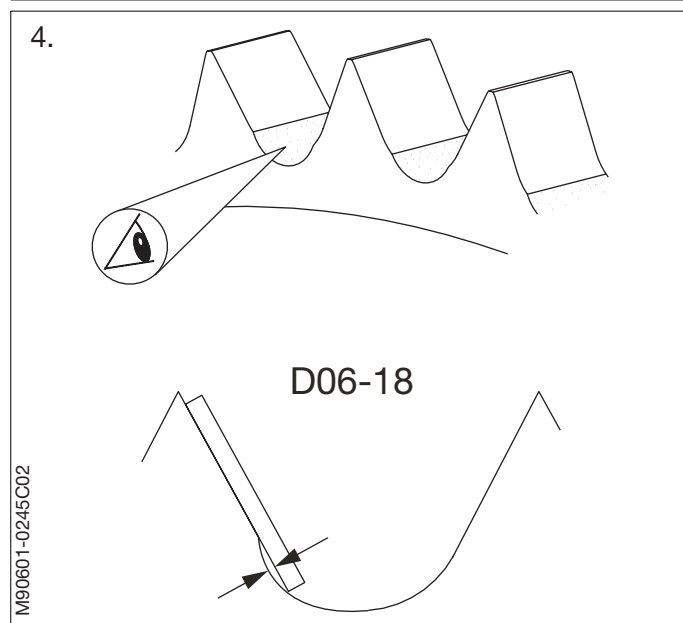
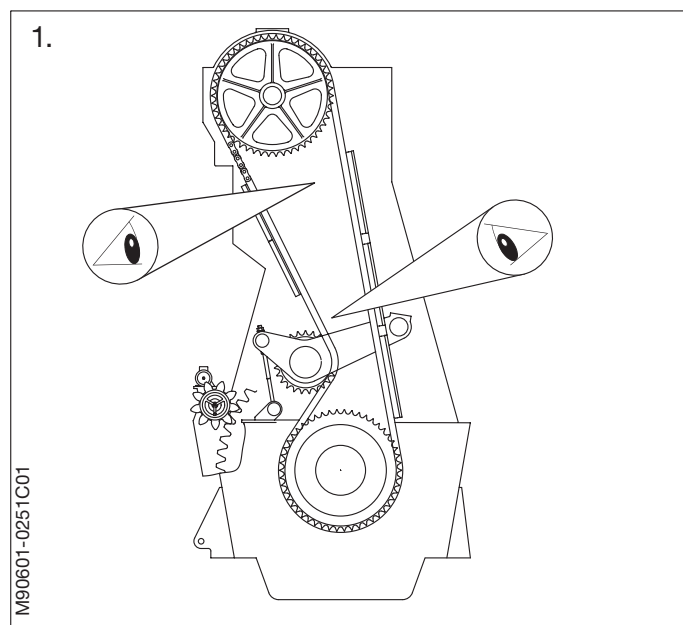
Measurements are best taken by placing a short straight-edge over straight part of the chain wheel and then measuring the distance D06-18.

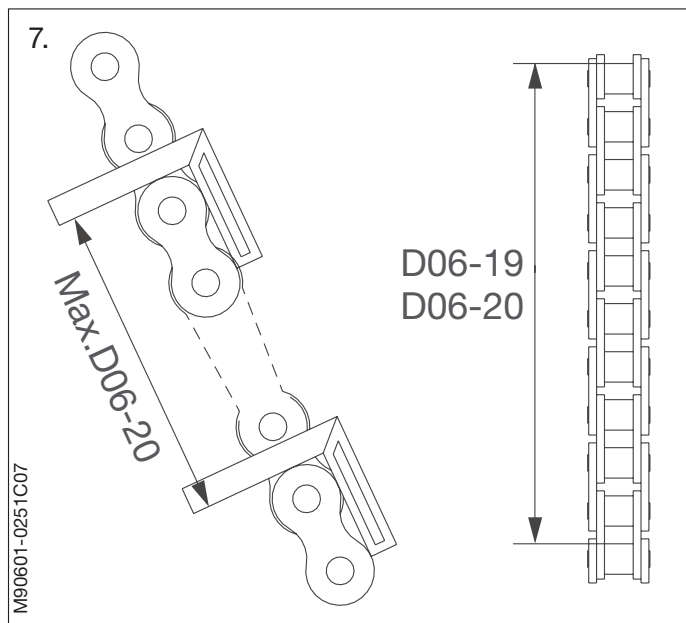
5. If abnormal wear is observed at the bottom of the teeth, make a drawing in scale 1:1 of the teeth and wear profile.

Draw two rulers on the drawing.

For assessing the wear profile, contact MAN Diesel.

In most cases, scratches caused by the side plates of the chain will be found on the sides of the teeth. Such scratches can generally be considered normal.





6. The chains for the camshaft drive are matched together to ensure an even load distribution.

To keep such matching chains in their pairs, the side plates of the outer link nearest to the assembled link have been marked with year, month, day and chain number.

Example:

No. 1 order,
840520 1A
840520 1B
(840520 1C, possible 3rd chain)

No. 2 order,
840520 2A
840520 2B
(840520 2C, possible 3rd chain)

On the same links there is an arrow (→) which indicates the mounting direction.

Check the chains for cracks on possibly defective rollers and side plates.

Check that the chain rollers can run freely and that the chain links can freely move on the pin and bushing (that they are not “seized” between the pin and the bushing).

It is normal, however, that the rollers get light, circumferential scratches during the running-in period. These fine scratches are of no importance and need not be considered.

It is recommended that each single link is checked.

7. Check chain wear by measuring the length of 10 chain links. Use two master squares and a steel measuring tape as shown on the sketch. Compare the result with the values given in Data.

If necessary, adjust the chain tightener.
See *Procedure 906-2.3*.

It may become necessary to disassemble the chain if, for instance, cracked rollers or seizures between pin and bushing have been discovered during the inspection.
See *Procedure 906-1.1*.

Note!

Every time a chain link is disassembled, a new link must **always** be fitted as the link pin press-fit is destroyed when breaking the chain.

When a new link is fitted in one chain, the corresponding link in the other chain must also be renewed.

1. Remove the tension on the chain by loosening the chain tightener.
(See *Procedure 906-2.3*).

Turn the engine until the slack part of the chain, with the chain link that is to be disassembled, is in a favourable position for the work. If the engine is equipped with balanceweights, continue the turning until the balanceweights are hanging vertically downwards, as shown in the sketch.

2. Mount a wire round the link rollers a short distance from the disassembly point, and tighten the wire lightly with a tackle.

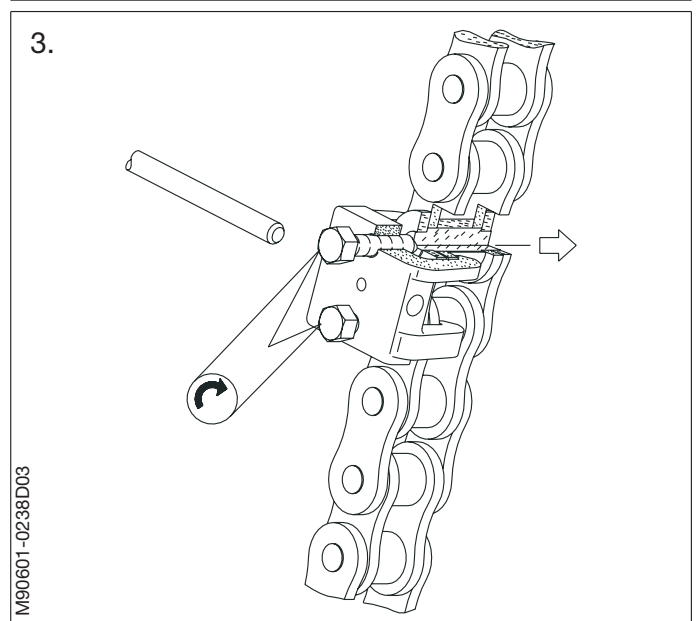
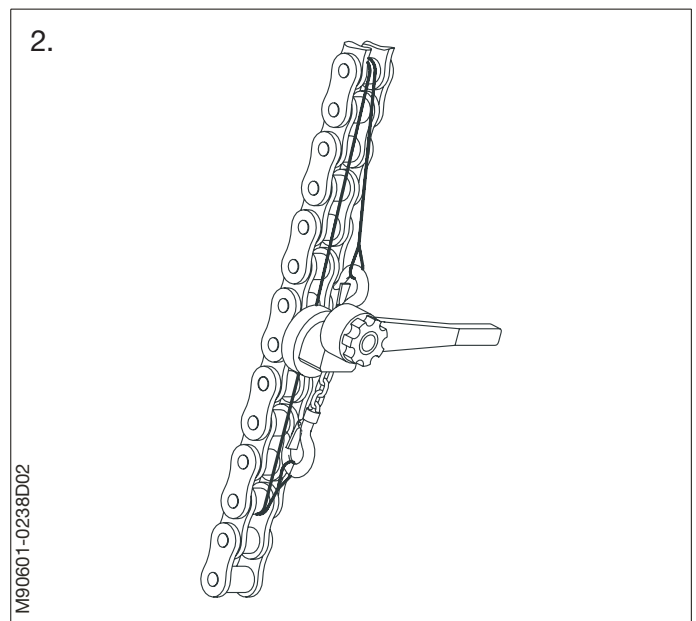
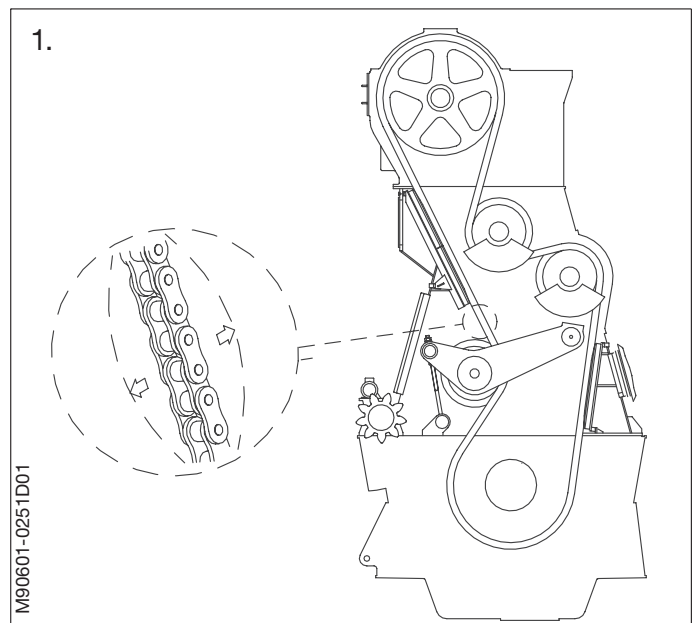
Protect the link rollers over which the wire is wrapped.

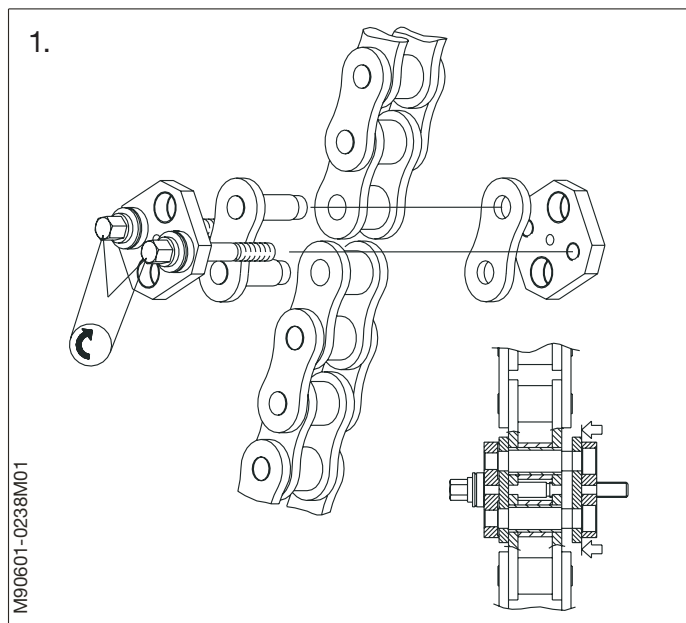
Note!

If the chain is to be completely removed, contact MAN Diesel for further information.

The riveting of the pins that are to be pressed out is to be chiselled or ground away.

3. Place the chain bursting tool over the outer chain link, and dismantle the link by alternately tightening the screws on the tool.





Before assembling the inner and outer links, clean the pins and bushings.

1. Combine the inner chain link with the outer chain link and mount the compression tool.

Force the loose side plate of the outer link into place by alternately tightening the screws on the compression tool.

2. When the link has been assembled, remove the compression tool and lock the pin ends by riveting.

Repeat this procedure until the chain has been assembled.

Remove the tackle and wire and adjust the chain tension.

(See *Data and Procedure 906-2.3*).

SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Shut off lubricating oil
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D06-21	Tightening angle = xx hexagons	720/12	° / hex
D06-23	Reduced tightening angle = xx hexagons	600/10	°
D06-25	Mechanical or blocked chain tightener, clearance	0.1	mm
D06-26	Max.distance	265	mm
D06-66	D max.	165	mm

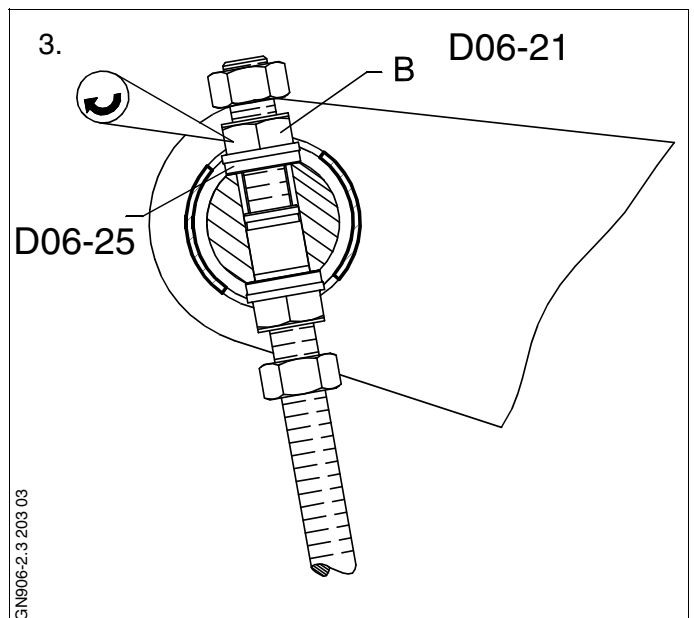
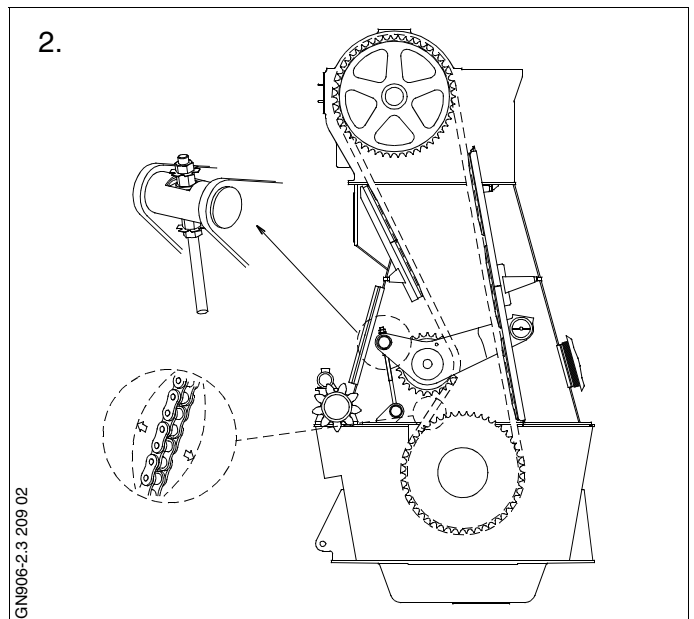
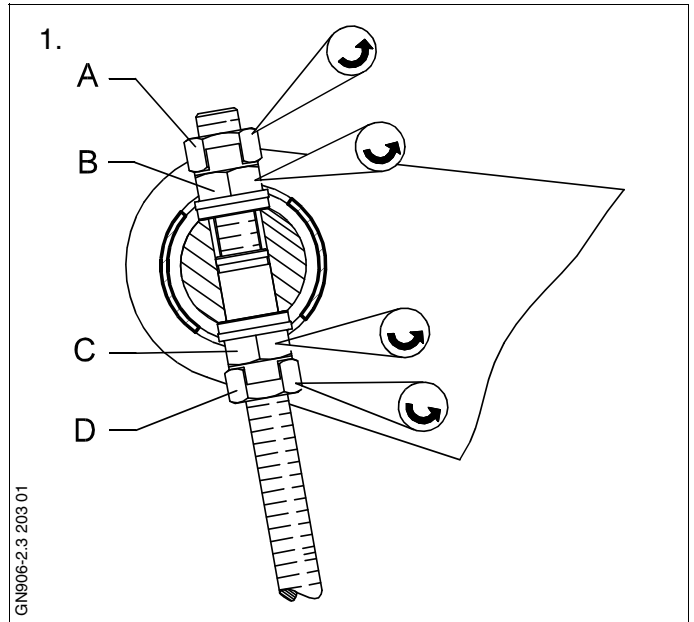
The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

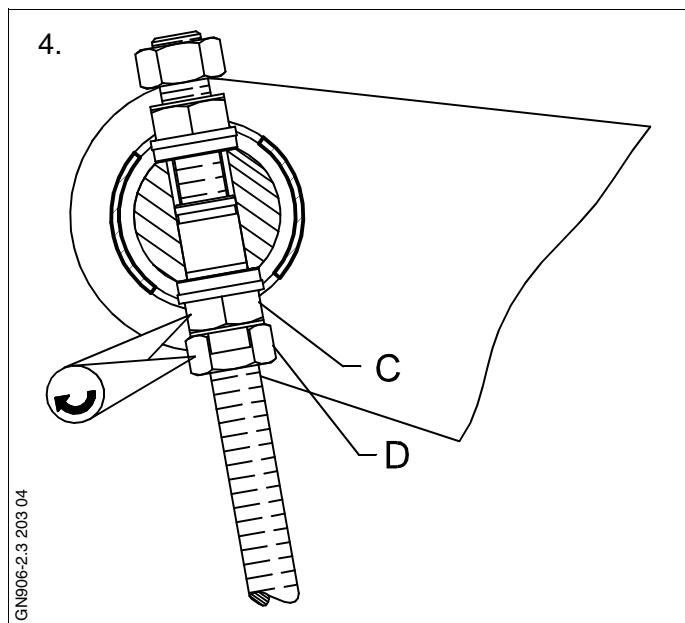
Plate	Item No.	Description
P91366	50	Feeler gauge

1. To retighten the chains, loosen nuts **A**, **B**, **C** and **D** to free the chain tightener bolt.
2. Turn the engine so that the slack part of the chains is on the same side as the tightener wheel.
3. Tighten nut **B** on the chain tightener bolt until there is a clearance between the shaft and the nut as stated in D06-25.

Then tighten nut **B** as stated in D06-21.
(see Data).

For chain wear, see Procedure 906-1.1.





4. Tighten nut **C** hard against the contact face of the shaft.

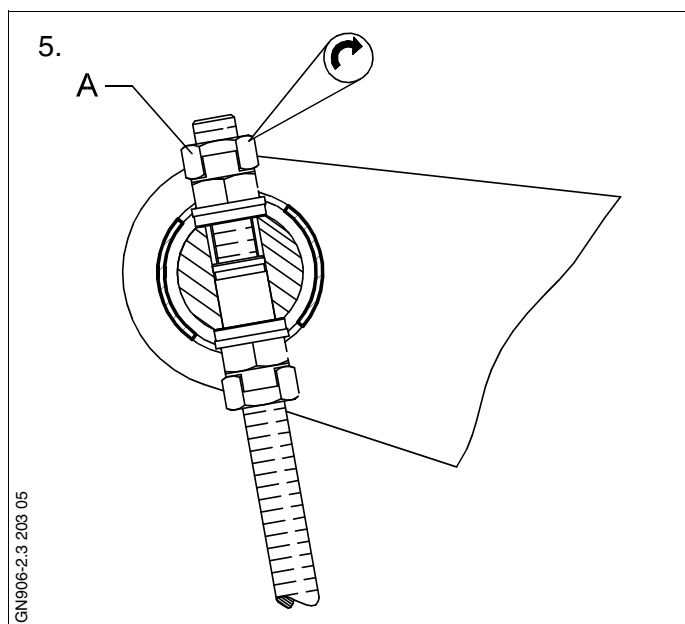
Tighten nut **D**.

Lock nuts **C** and **D** with the tab washer.

5. Tighten nut **A**.

Lock nuts **A** and **B** with the tab washer.

The number of times that retightening can be carried out depends on when the limit for scrapping the chain has been reached. See *Data page 106-1*.



SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☐	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D06-27	Max. permissible change in lead angle	2	°
D06-28	corresponding to	3.5	mm

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P90662 P90664 P91366	061	Camshaft - Hydraulic Tools Crankshaft - Pin Gauge Slide caliper

1. Turn the crankthrow for cylinder 1 to TDC.

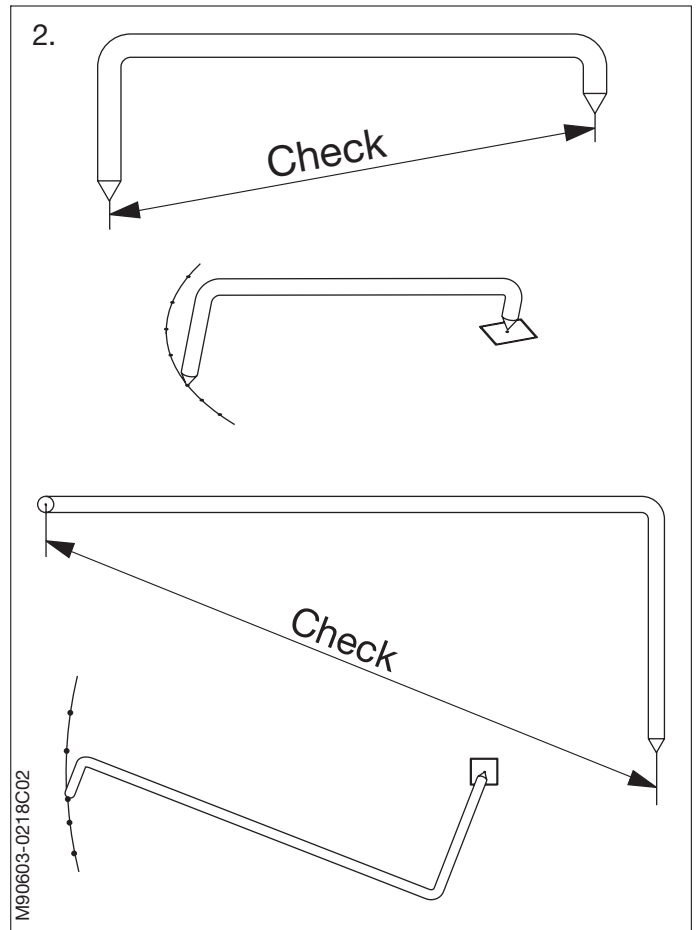
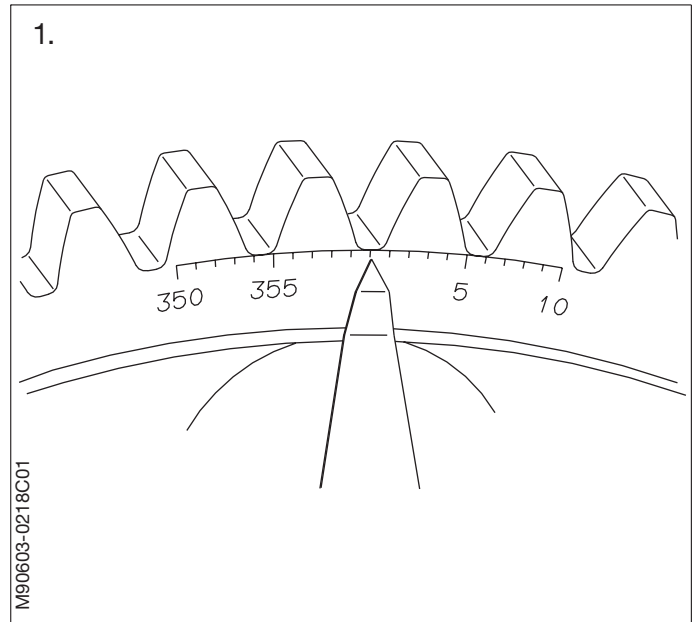
Turn in the ahead direction to ensure that the tension of the chain on the chain drive is correct. If necessary, adjust the chain tension.

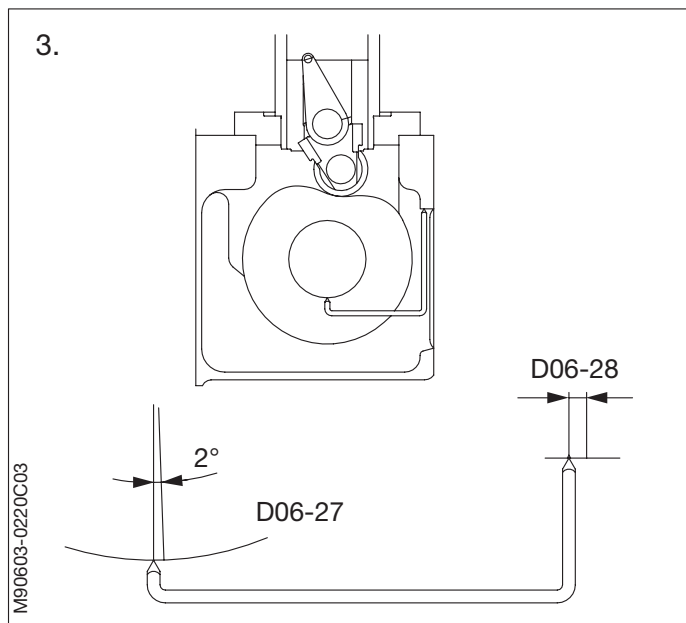
See *Procedure 906-2.3*.

Check the TDC of cylinder 1 against the mark on the turning wheel.

2. Use the check measurement value stamped on the pin gauges to check the pin gauges for deformation.

On some engines the pin gauges can be checked against punch marks made on the front cylinder frame.





3. Position the camshaft pin gauge in the centre punch mark, and measure the deviation, if any, in lead angle.

For max. permissible change, see Data.

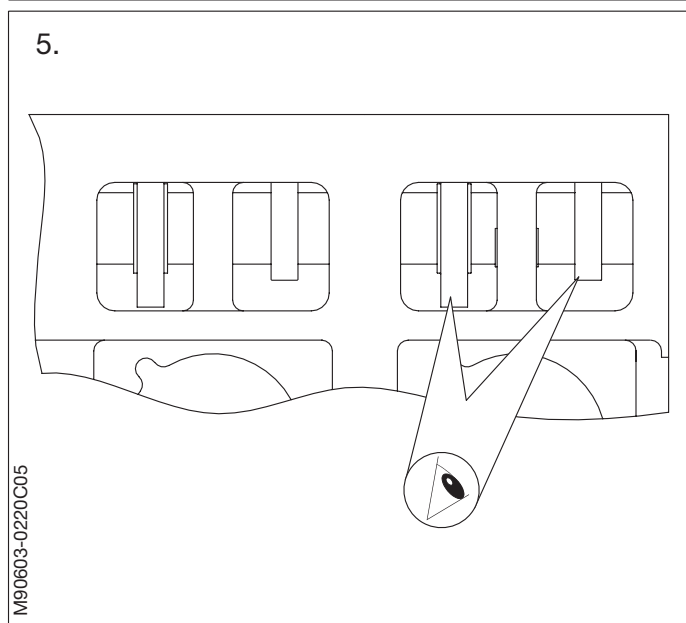
4. Check and, if necessary adjust the cylinder lubricator. See Procedure 903-2.1.

Check the setting of the starting air distributor. See Procedure 907-1.1.

5. When checking the camshaft position, also check the running surface of the cams.

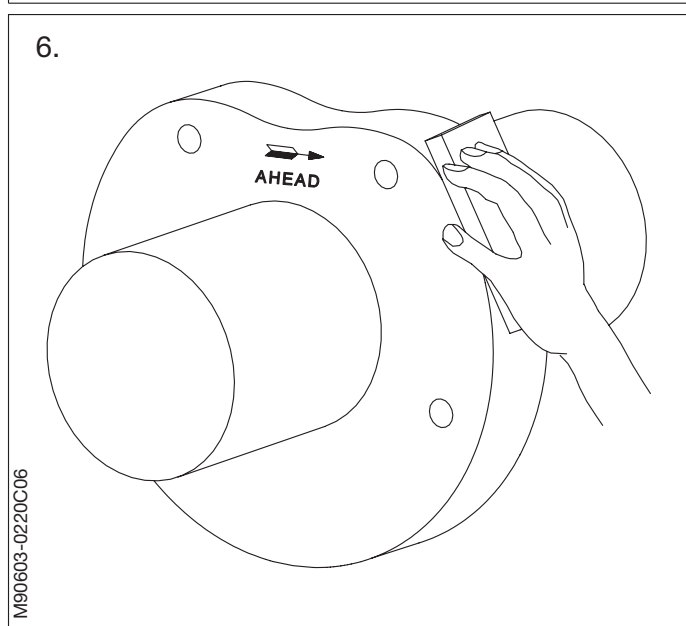
As the inspection must include the entire surface of the cam, the cam should be turned one complete revolution during the inspection.

The running surface of the cam must be completely smooth and bright.

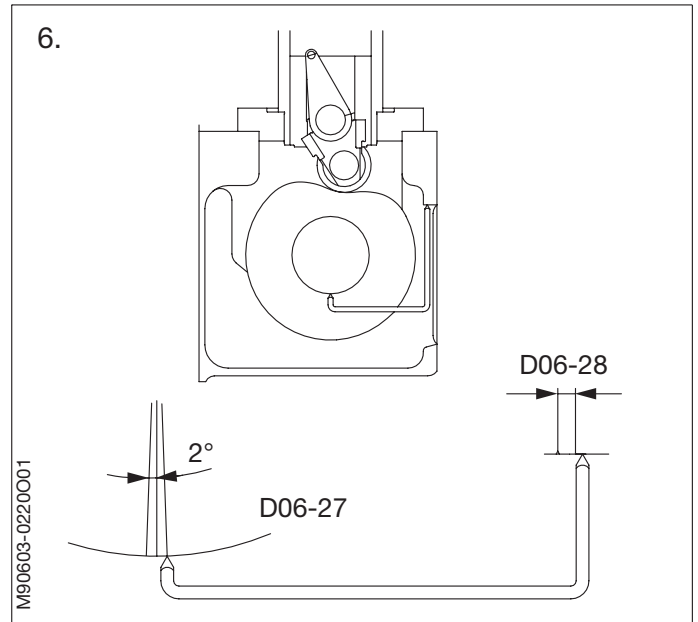


6. If light scratches are ascertained, remove them by carefully polishing the cam in the direction of rotation with a fine carborundum stone. Such light scratches can appear if the roller guide and the cam are not correctly aligned.

In the event of heavier scratches or direct flaking of the material on the running path, MAN Diesel should be contacted for further instructions.



1. If the chains prove to be so heavily worn that the deviation between the measuring points and the pin gauge has reached the maximum value stated in *Procedure 906-3.1*, MAN Diesel should be contacted for further instructions.



SAFETY PRECAUTIONS *For detailed sketch, see 900-2*

☒	Stopped engine
☒	Shut off starting air supply – <i>At starting air receiver</i>
☒	Block the main starting valve
☒	Shut off starting air distributor/distributing system supply
☒	Shut off safety air supply – <i>Not ME engines</i>
☒	Shut off control air supply
☒	Shut off air supply to exhaust valve – <i>Only with stopped lubricating oil pumps</i>
☒	Engage turning gear
☐	Shut off cooling water
☐	Shut off fuel oil
☒	Stop lubricating oil supply
☐	Lock the turbocharger rotors

Data

Ref.	Description	Value	Unit
D06-31	Screws for bearing cap, tightening torque	125	Nm
D06-33	Clearance between cap/bearing housing	3 ± 0.3	mm
D06-34	Bearing shell, max. wear	0.1	mm
D06-37	Camshaft bearing cap	13	kg
D06-65	Bearing shell, minimum thickness	7.35	mm

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. **P90951** refers to chapter 909.

Plate	Item No.	Description
P91351	022	Hydraulic pump, hand operated
P91351	058	Hose with unions (3000 mm), complete
P90651	152	Hydraulic jack for lift of camshaft
P91356		Lifting Tools, Etc.
P91359		Torque Spanners
P91366	061	Slide caliper

1. As the upper shell of the camshaft bearing assembly has been dispensed with on this engine, it is no longer necessary to measure the top clearance of the camshaft bearing.

Dismount the inspection covers from the front of the camshaft housing.

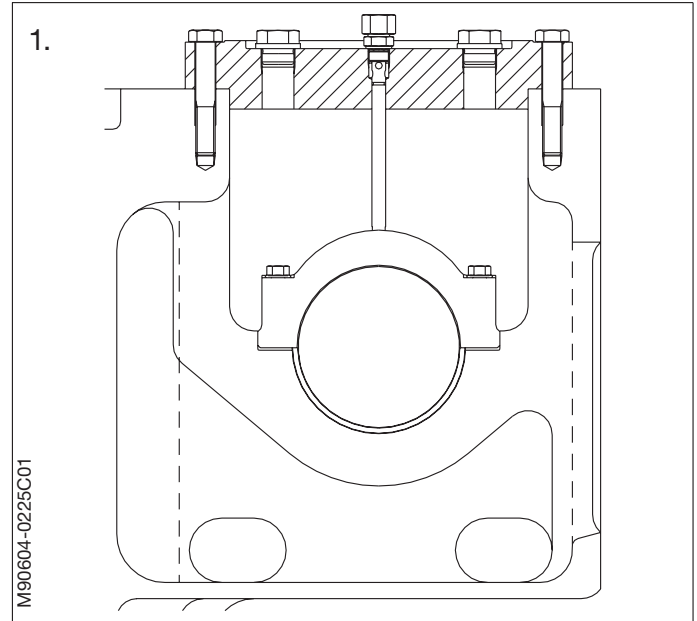
Check for wiped-out metal at the bearing ends, and for fragments of metal in the camshaft housing.

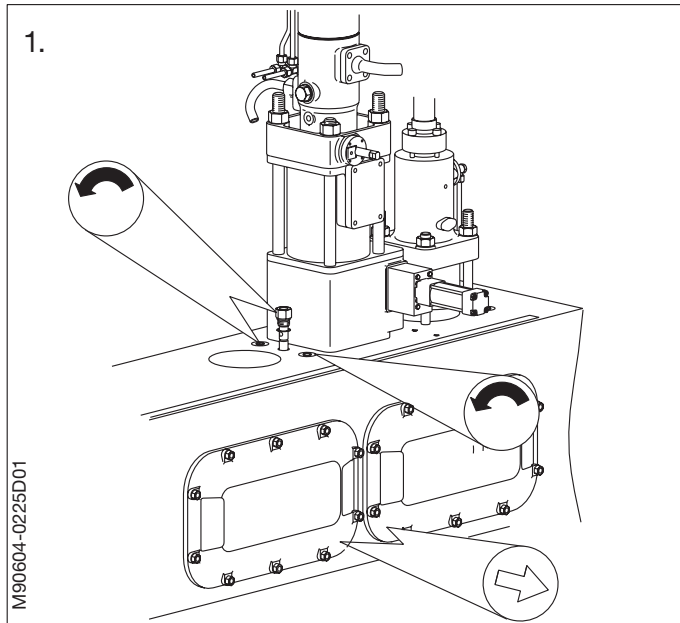
In the event of any signs of anomalies, the bearing shell must be dismantled for closer investigation and, if necessary, the bearing shell must be replaced.

If no anomalies are found during the visual inspection, dismantling need not be carried out.

When checking the camshaft bearings, it is recommended also to check the condition of the fuel and exhaust cams.

See *Procedures 908-5 and 909-3*.



**Warning!**

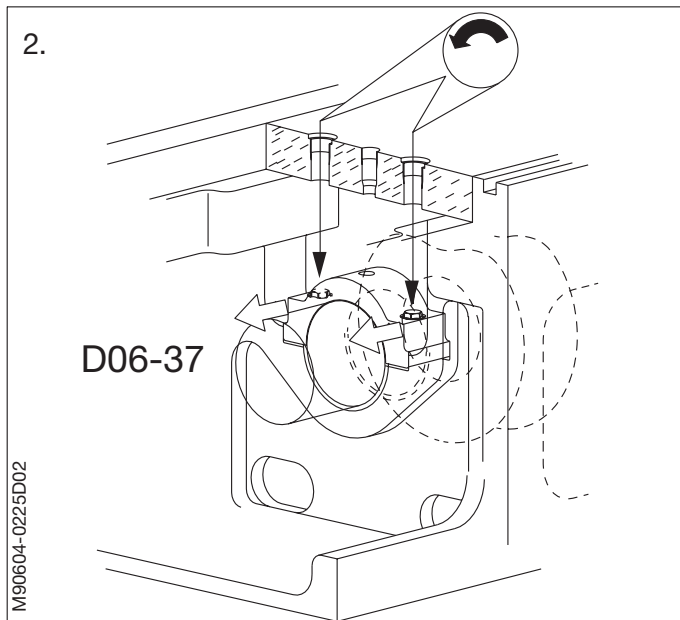
Before dismantling any camshaft bearings, the main chain **must** be relieved of tension by loosening the chain tightener to avoid damaging engine parts.

See Procedure 906-2.3.

1. Remove the inspection covers on each side of the bearing from the camshaft housing.

Dismount the lubricating oil connecting pipe and the plug screws from the base-plate above the bearing in question.

Lift the roller guide of the exhaust valve actuator. See Procedure 908-7.



2. Loosen the bearing cap screws through the holes in the baseplate.

Remove the screws from the bearing cap.

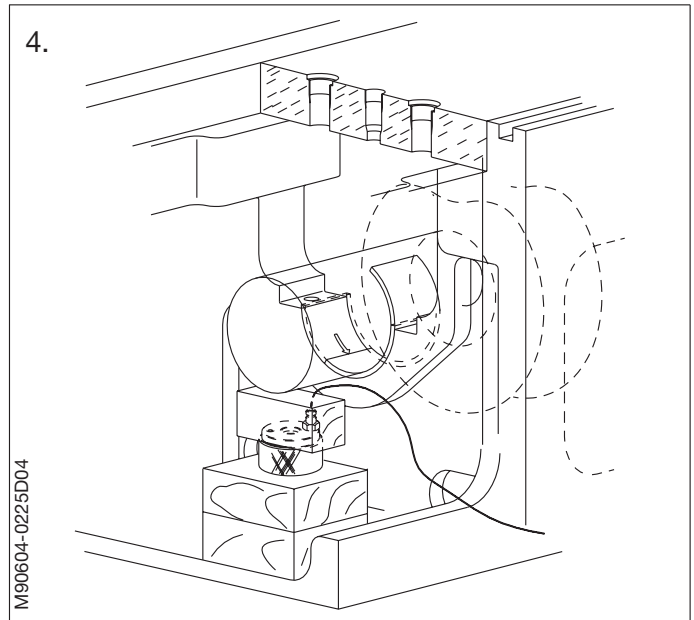
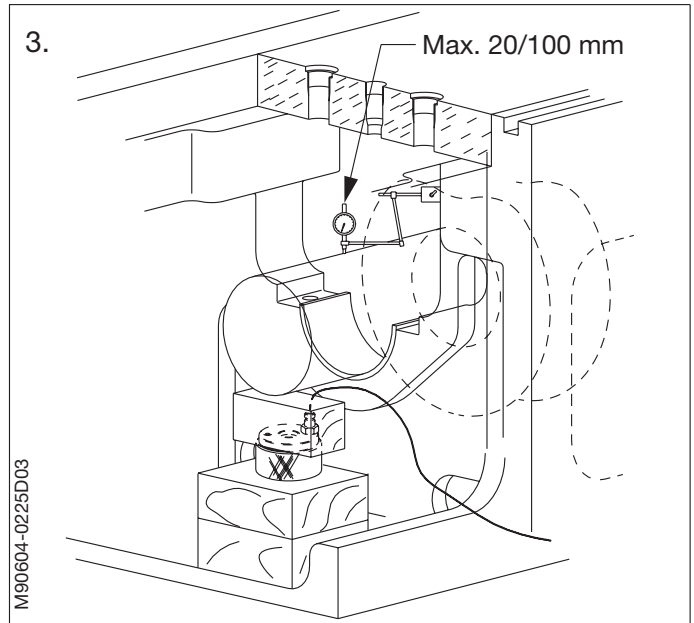
Remove the bearing cap from the housing.

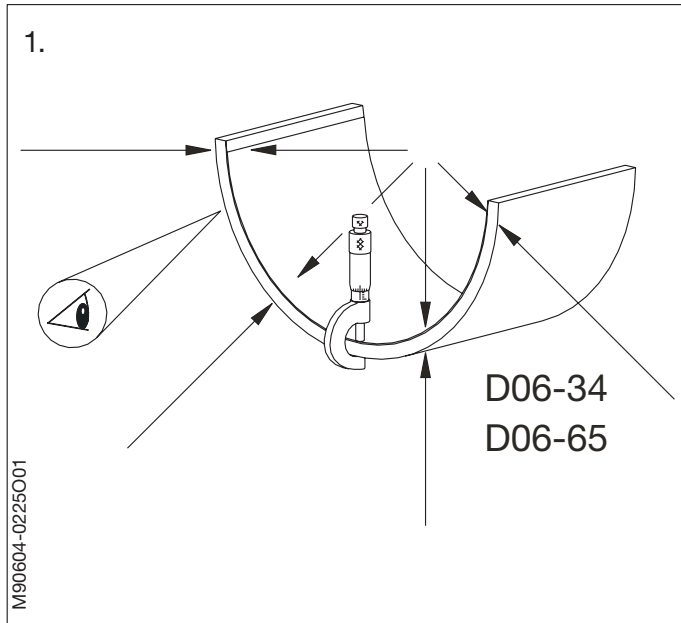
3. Place a hydraulic jack between two pieces of wood in the camshaft housing underneath the camshaft, close to the bearing.

Place a dial gauge on the camshaft, and lift the camshaft 0.20 mm.

4. Press lightly on the upper edge of the shell to turn the shell up over the camshaft.

Remove the shell from the camshaft housing.





1. For assessment of the bearing condition, see *Volume I OPERATION, Chapter 708 'Bearings'*.

Inspect the entire surface of the bearing journal for seizures.

If seizures are found on the bearing journal surface (on account of bearing breakdown), polish the surface with a fine carborundum stone to re-establish a bright and smooth surface.

By closely examining the sliding surface of the bearing shell visually, an impression can be formed of whether the lubrication of the bearing surface has been insufficient or entirely missing. Normally, camshaft bearing damage only occurs due to faulty lubrication.

Measure the thickness of the bearing shell with a micrometer to check the amount of wear.

1. Clean the bearing journal, the bearing support and the bearing shell.
2. Coat the bearing journal and the bearing shell with plenty of lubricating oil. Mount the shell on the camshaft, and turn it into place.

Relieve the hydraulic jack of pressure to lower the camshaft. Remove the dial gauge, the pieces of wood and the hydraulic jack from the camshaft housing.

3. Place the bearing cap on the camshaft and mount the outermost bearing cap screw.

Note!

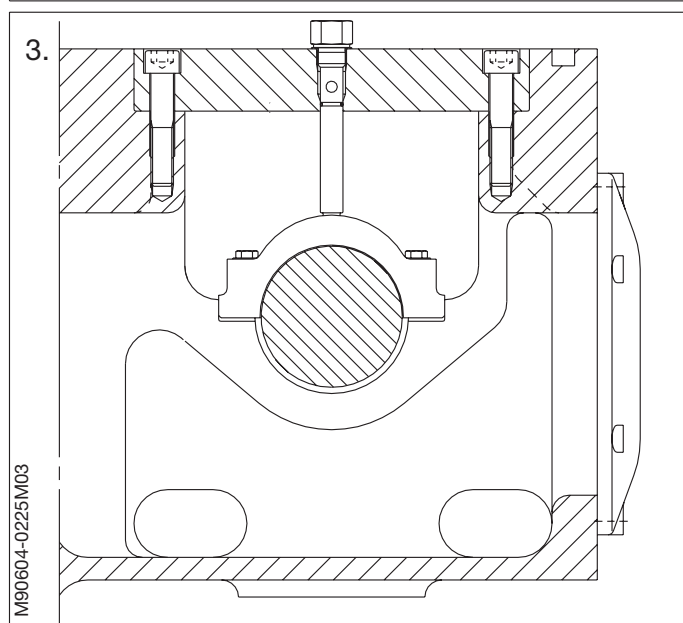
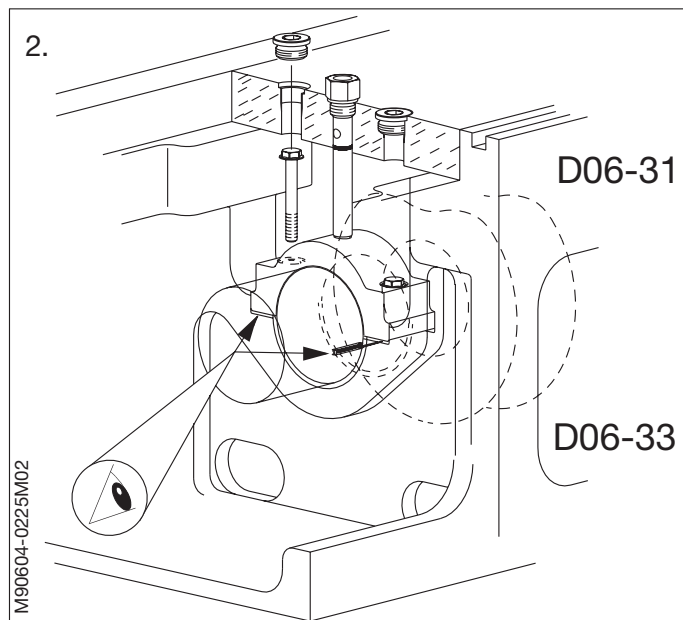
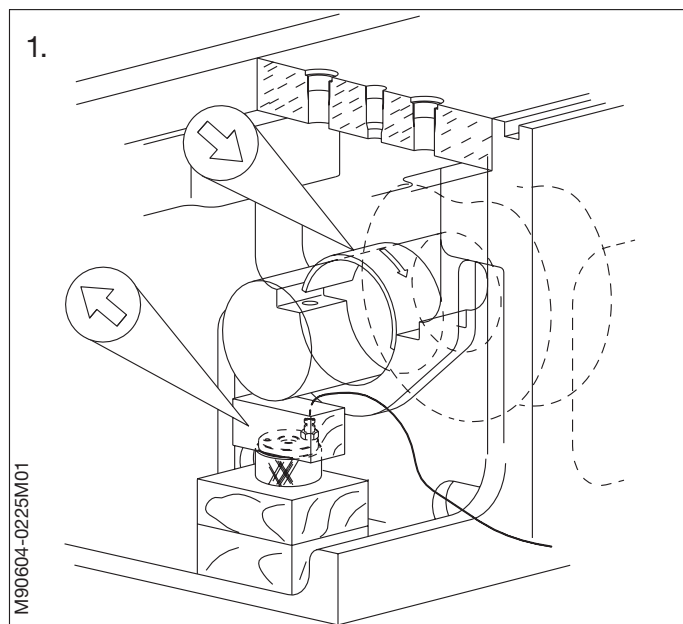
Be careful not to scratch the camshaft surface.

Tighten the outermost screw lightly by hand until the clearance given in D06-33 is reached. Then mount and tighten the innermost screw with a torque spanner. See *Data D06-31*. This procedure ensures that the outermost screw will obtain the correct torque.

Check the torque on the outermost screw with a torque spanner.

Check the clearance D06-33 on both sides of the camshaft. Then mount the lubricating oil pipe and the plug screws in the base-plate.

4. Lower the exhaust valve actuator roller guide. See *Procedure 908-7*.
5. Clean the camshaft housing and mount the inspection covers.



SAFETY PRECAUTIONS

☒	Stopped engine
☒	Block the starting mechanism
☒	Shut off starting air supply
☒	Engage turning gear
☐	Shut off cooling water
☒	Shut off fuel oil
☒	Shut off lubricating oil
☐	Lock turbocharger rotors

Data

Ref.	Description	Value	Unit
D06-47	Exhaust cam screws, tightening torque	560	Nm
D06-48	Indicator cam screws, tightening torque	50	Nm
D06-49	Fuel cam screws, tightening torque	880	Nm
D06-50	Fuel cam, marking scratch	65	mm
D06-51	Fuel cam, axial movement	145	mm
D06-52	Exhaust cam, marking scratch	55	mm
D06-53	Exhaust cam, axial movement	220	mm

The task-specific tools used in this procedure are shown on the plates at the end of this chapter or in the chapters indicated by the first three digits in the plate number, e.g. P90951 refers to chapter 909.

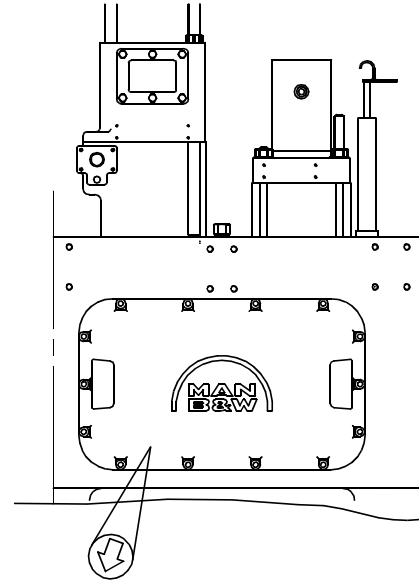
Plate	Item No.	Description
P91351	22	Hydraulic pump, hand operated
P91351	58	Hose with unions (3000 mm), complete
P91351	105	3-way distributor block, complete
P91359		Torque spanners

Fuel Cam

1. Remove the front cover of the camshaft housing.
2. Lift the roller guide from the cam in question. See *Procedures 908-5 or 909-5*.
3. Turn the camshaft so that the marking scratch can be seen.
4. Extend the marking scratch on the shaft by D06-50. See *Data*.

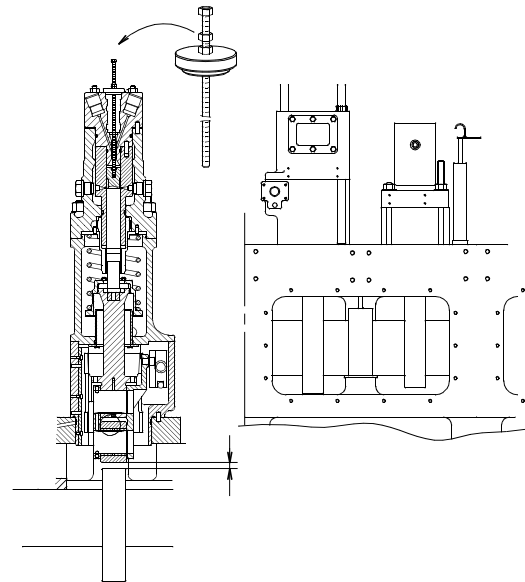
1.

XZ906-6.2 201 01



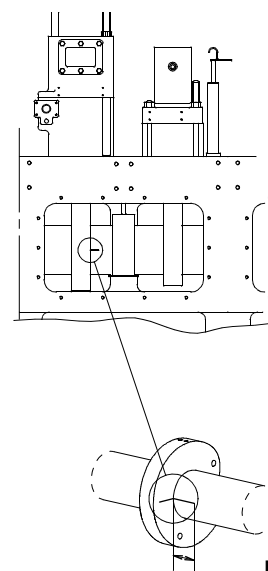
2.

XZ906-6.2 201 02



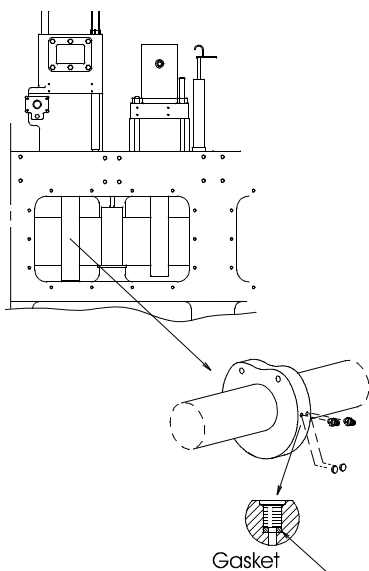
4.

XZ906-6.2 201 04



D06-50

5.



5. Remove the plug from the cam and mount gaskets and snap-on connectors instead.

6. Connect the snap-on connectors to the hydraulic high-pressure pump.

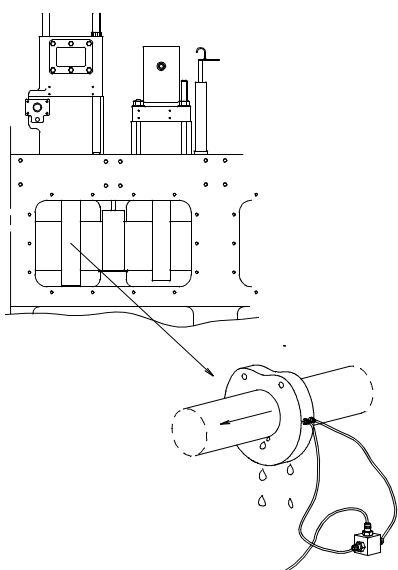
Apply oil pressure to the cam, using the high-pressure pump.

When oil seeps out along the cam, press the cam sideways, in the direction towards the forward end of the engine.

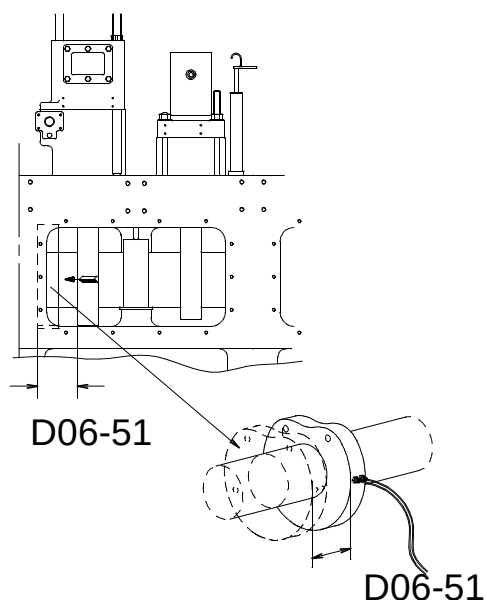
7. The cam should be moved as stated in D06-51 to be sure that it gets free of the liner for the roller guide.

8. In case the cam cannot be moved as described in item 6, MAN B&W should be contacted for further instructions.

6.



7.



XZ906-6.2 201 05

XZ906-6.2 201 06

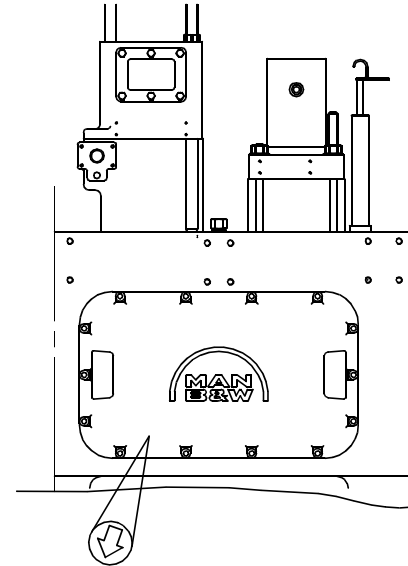
XZ906-6.2 201 07

Exhaust Cam

9. Remove the front cover of the camshaft housing.
10. Lift the roller guide from the cam in question. See *Procedure 908-5*.
11. Extend the marking scratch on the indicator drive cam onto the camshaft.

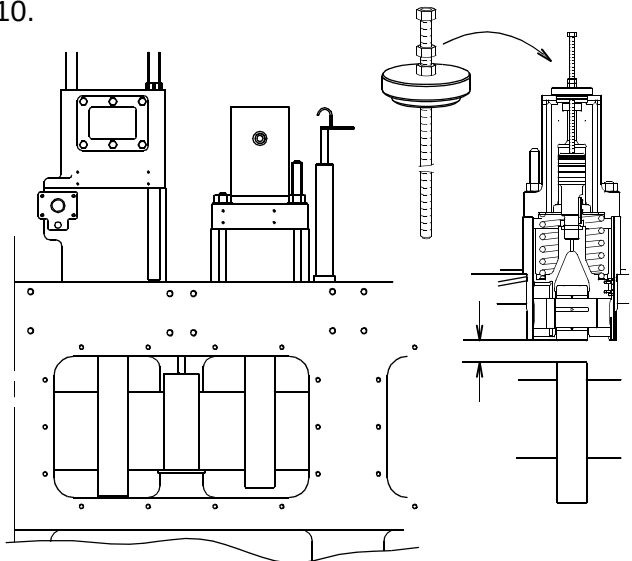
Extend the marking scratch on the exhaust cam onto the camshaft by D06-52. See *Data*.

9.



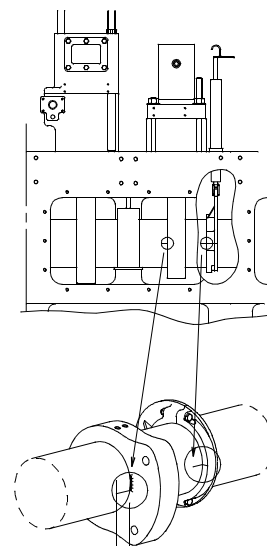
XZ906-6.2 201 01

10.



XZ906-6.2 201 10

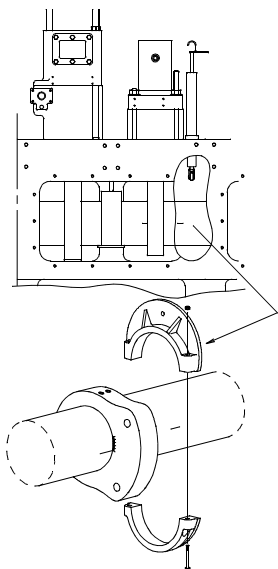
11.



XZ906-6.2 201 11

D06-52

12.



12. Dismantle and remove the indicator drive cam from the camshaft when fitted.

13. Remove the plug from the cam and mount gaskets and snap-on connectors instead.

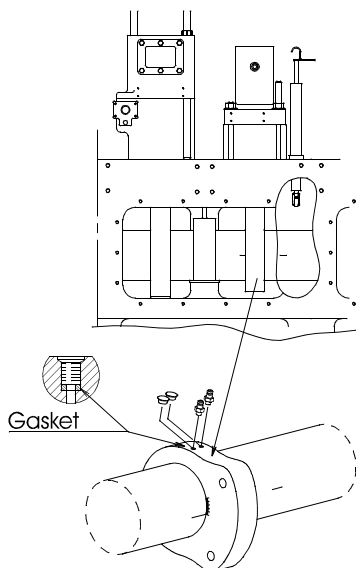
14. Connect the snap-on connectors to the hydraulic high-pressure pump.

Apply oil pressure to the cam, using the high-pressure pump.

When oil seeps out along the cam, press the cam sideways, in the direction towards the aft end of the engine.

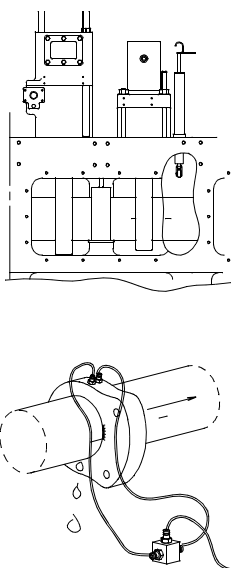
XZ906-6.2 201.12

13.



XZ906-6.2 201.13

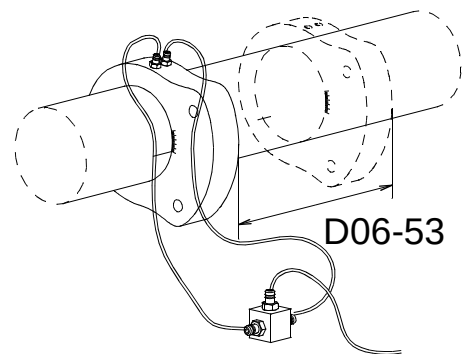
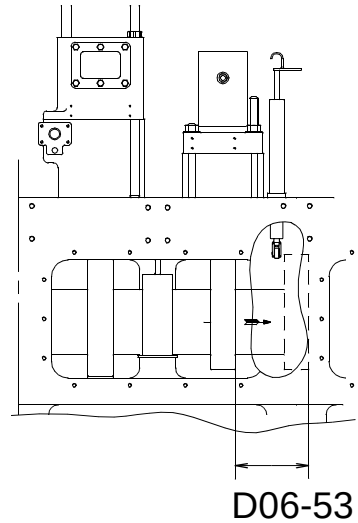
14.



XZ906-6.2 201.14

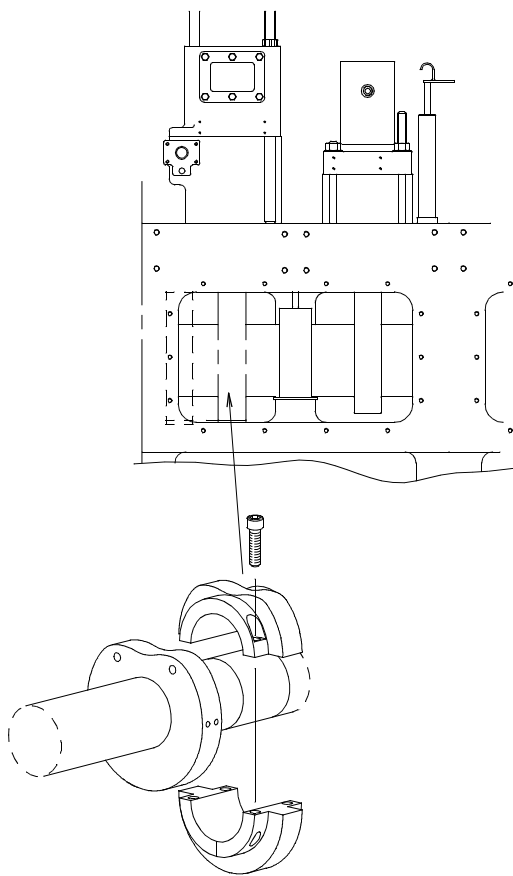
15. The cam should be moved as stated in D06-53 to be sure that it gets free of the liner for the roller guide.
16. In the event that the cam cannot be moved as described in item 14, MAN B&W should be contacted for further instructions.

15.



XZ906-6.2 201 15

1.

**Fuel Cam**

1. Before mounting the two-part spare cam, the bore and the area on the shaft must be degreased.

Place the two-part spare cam on the shaft, mount the four screws, and tighten slightly, so that the gap between the two parts is equal at both sides.

2. Make sure that the marking scratches on the cam and the camshaft coincide, and that the running surface of the cam is opposite the roller.

Tighten the screws just hard enough to avoid that the cam slips when the roller guide is lowered onto the cam.

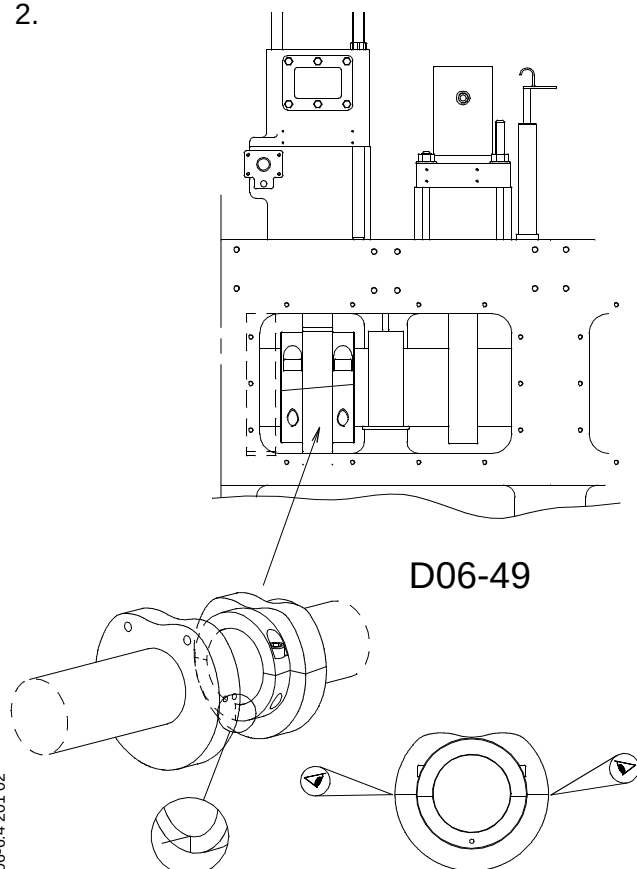
Measure the fuel cam lead (see *Procedures 909-1 and 909-2*) and adjust if necessary.

Tighten the screws to D06-49 and check that the gap is still equal at both sides.

Lock the screws with Loctite, type 242 (EN243S).

3. Mount the cover on the front of the camshaft housing.

2.



D06-49

Exhaust Cam

4. Mount the indicator drive cam and adjust according to Chapter 706 in the Operation manual when fitted.

Tighten the self-locking nuts as stated in D06-48.

5. Before mounting the two-part spare cam, the bore and the area on the shaft must be degreased.

Place the two-part spare cam on the shaft, mount the four screws, and tighten slightly, so that the gap between the two parts is equal at both sides.

6. Make sure that the marking scratches on the cam and the camshaft coincide, and that the running surface of the cam is opposite the roller.

Tighten the screws just hard enough to avoid that the cam slips when the roller guide is lowered onto the cam.

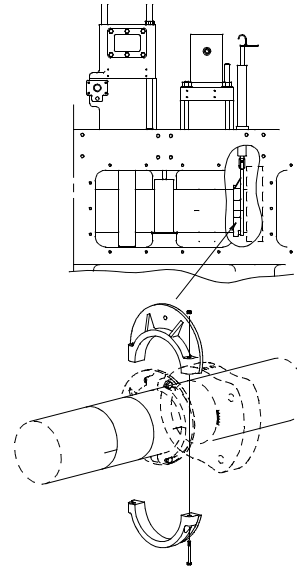
Measure the fuel cam lead (see *Procedure 908-7*) and adjust if necessary.

Tighten the screws to D06-47 and check that the gap is still equal at both sides.

Lock the screws with Loctite, type 242 (EN243S).

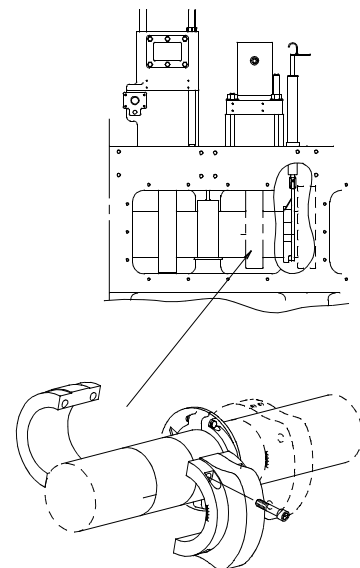
7. Mount the cover on the front of the camshaft housing.

4.



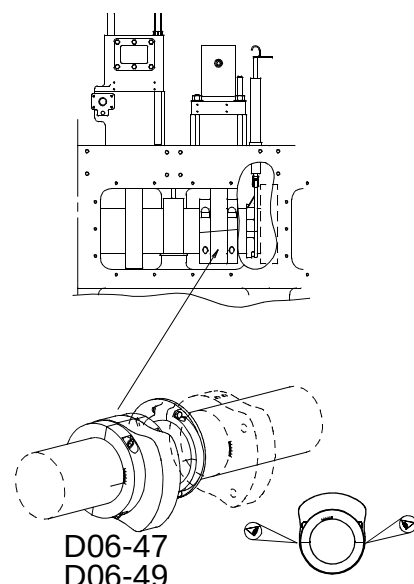
XZ906-6.4 201 04

5.

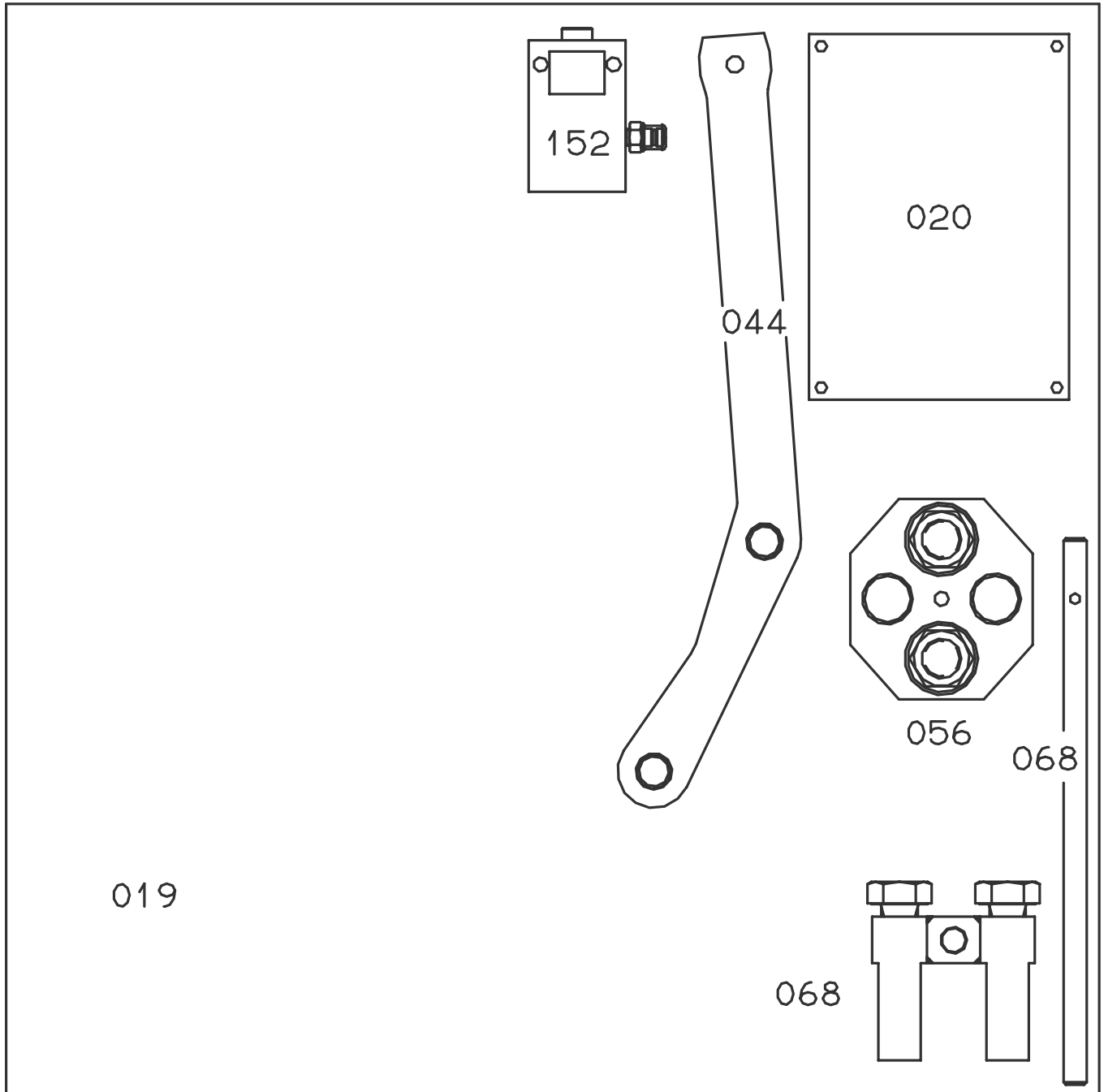


XZ906-6.4 201 05

6.



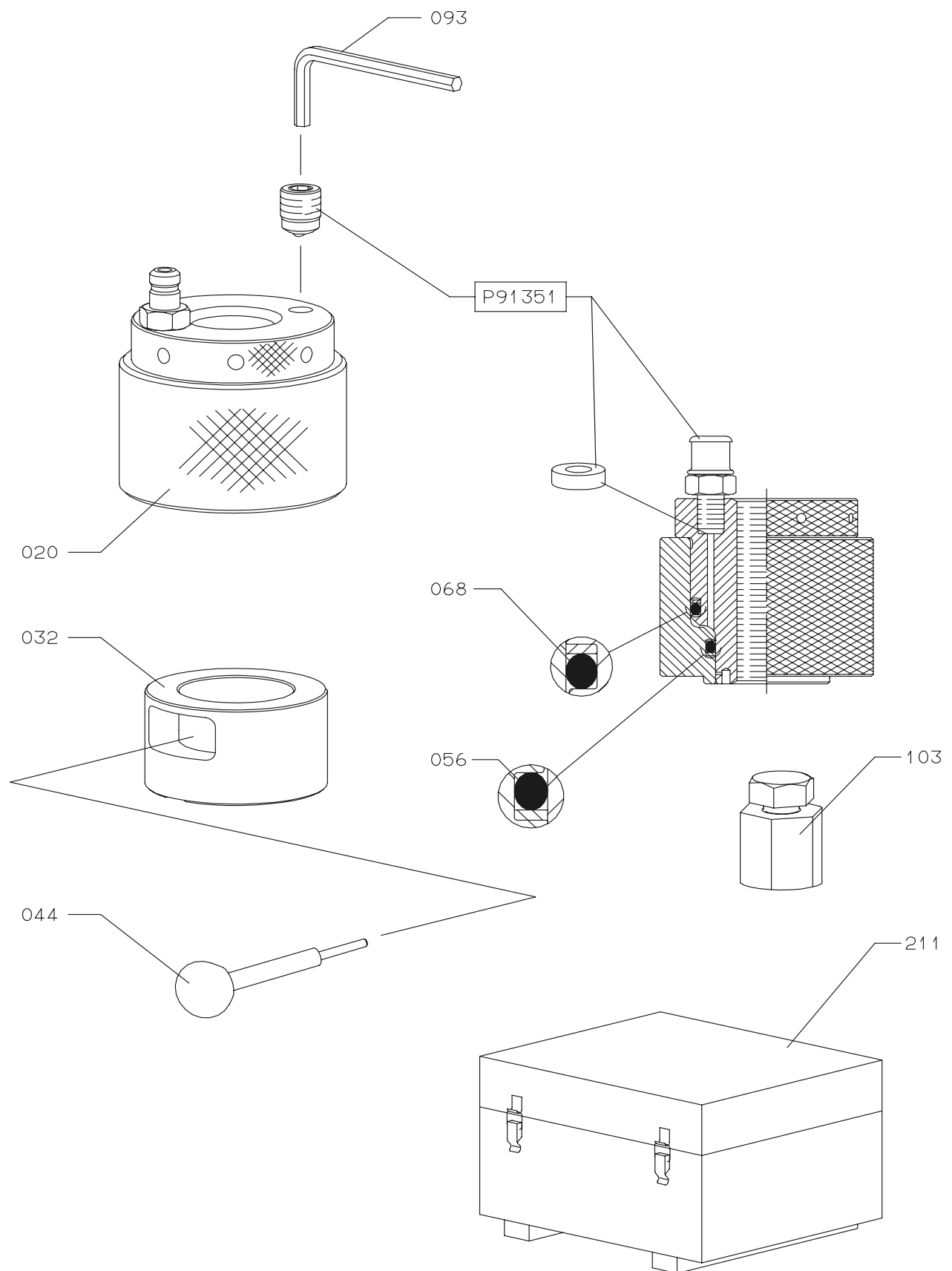
XZ906-6.4 201 06



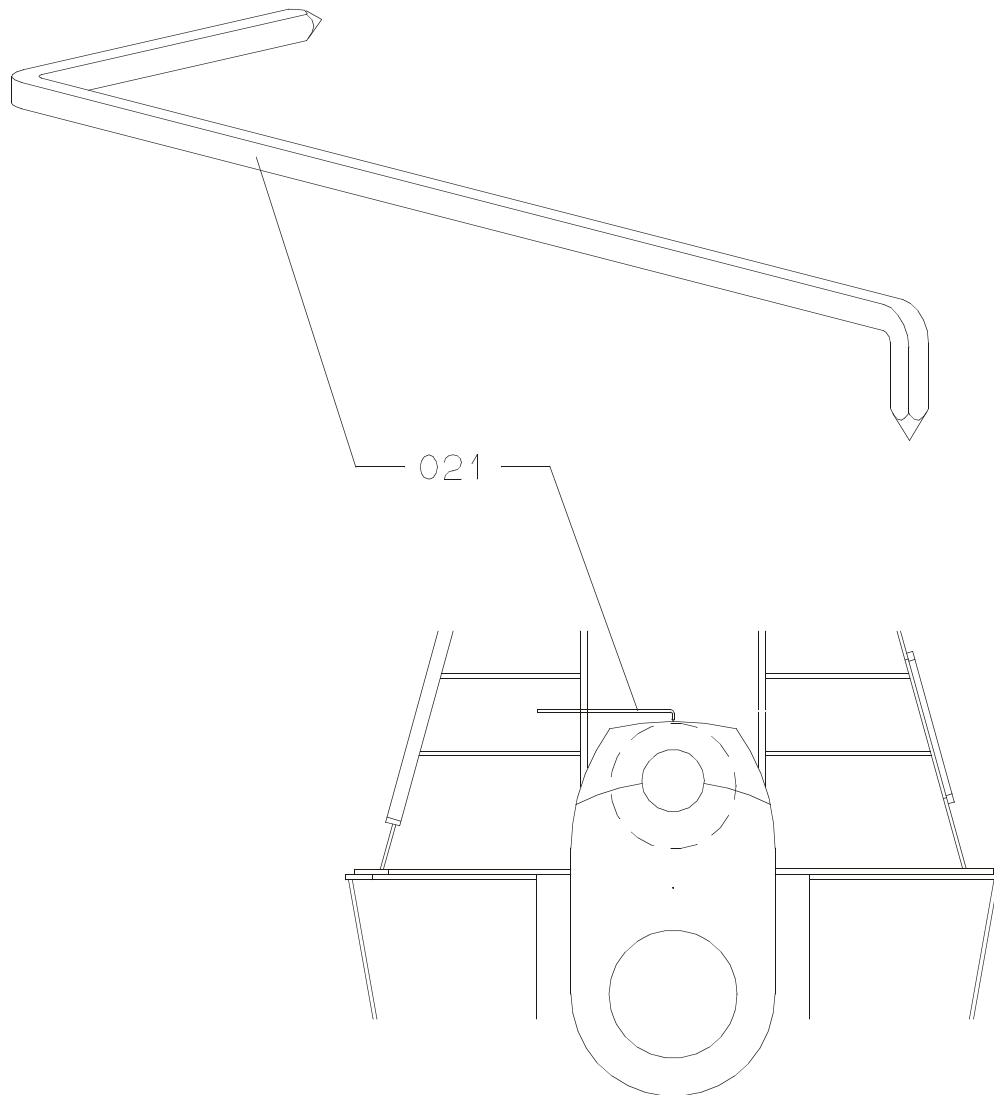


Item No.	Item Description
019	Panel for tools
020	Name plate
044	Impact spanner, fuel cam
056	Assembling tool, chain
068	Disassembling tool, chain
152	Hydraulic jack, lift of camshaft

Item No.	Item Description

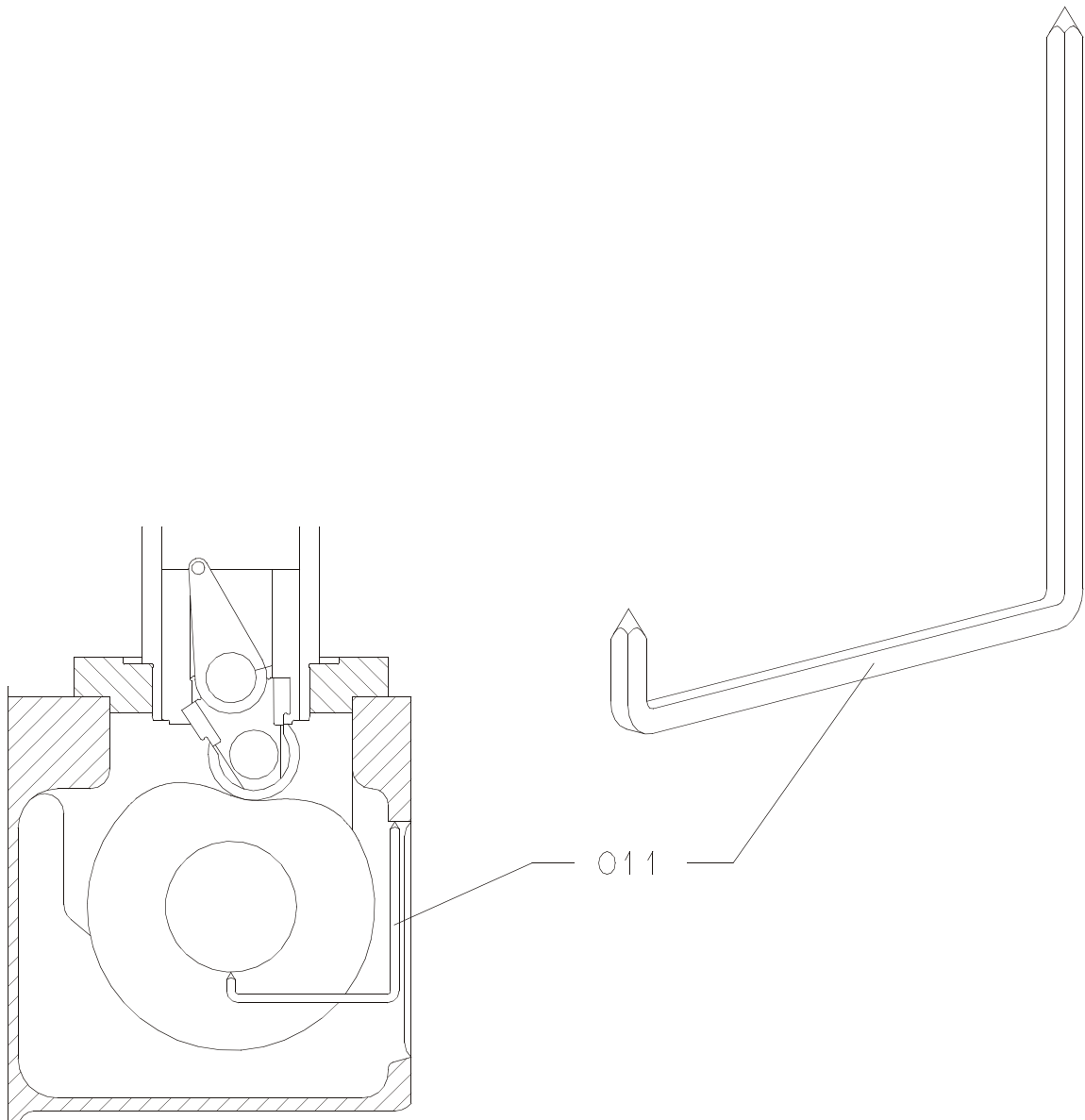


Item No.	Item Description



Item No.	Item Description
021	Pin gauge for crankshaft

Item No.	Item Description



Item No.	Item Description
011	Pin gauge for camshaft

Item No.	Item Description